

1.

2. He should be concerned about being overly exposed to Fin sector as this makes up $> 50\%$.

UNIVERSITY OF CAPE TOWN
Investment Decision Theory and Application
ECO4102Z /ECO4122Z
Passive Management: Activity 8

1. Given the monthly returns that follow, how well did the passive portfolio track the S&P 500 benchmark? Find the R^2 , α , and β of the portfolio. Compute the average return differential with and without sign. Calculate the tracking error of the portfolio

Month	R_p Portfolio Return	R_m S&P 500 Return
January	5.0%	5.2%
February	-2.3	-3.0
March	-1.8	-1.6
April	2.2	1.9
May	0.4	0.1
June	-0.8	-0.5
July	0.0	0.2
August	1.5	1.6
September	-0.3	-0.1
October	-3.7	-4.0
November	2.4	2.0
December	0.3	0.2

2. Axola is considering investing in a fund that tracks the MSCI Emerging Frontier Markets Africa ^{EFM ex} ex South Africa Index. He has obtained the MSCI Fund Fact Sheet (see attachment) which outlines the top 10 constituents and the sector weightings as of 30 August 2024. Explain whether Axola should be concerned about being overly exposed to a certain stock or sector, when investing in this index.
3. Global Advisers Company (GAC) is an SEC-registered investment counseling firm involved solely in managing international securities portfolios. After much research on the developing economy and capital markets of the country of Otunia, GAC has decided to include an investment in the Otunia stock market in its Emerging Market Commingled Fund. However, GAC has not yet decided whether to invest actively or by indexing. Your opinion on the active versus indexing decision has been solicited. A summary of the research findings follows.

Otunia's economy is fairly well diversified across agricultural and natural resources, manufacturing (both consumer and durable goods), and a growing finance sector. Transaction costs in securities markets are relatively large in Otunia because of high commissions and government "stamp taxes" on securities trades. Accounting standards and disclosure regulations are quite detailed, resulting in wide public availability of reliable information about companies' financial performance. Capital flows into and out of Otunia and foreign ownership of Otunia securities are strictly regulated by an agency of the national government. The settlement procedures under these ownership rules often cause long delays in settling trades made by nonresidents. Senior finance officials in the government are working to deregulate capital flows and foreign ownership, but GAC's political consultant believes that isolationist sentiment may prevent much real progress in the short run.

3. Active

1 - Growing finance sector
- this means there is an opportunity to

2 - Not fully diversified
-

Passive (Index)

1 - economy is well diversified
∴ the market may be efficient

2 - High transaction costs
- this may offset active returns

3 - Stamp taxes on securities trade
- Taxes may offset active returns

4 - Info is publically available + reliable
- This means there is less opportunity to act info asymmetry

- a. Briefly discuss four aspects of the Otunia environment that favor investing actively and four aspects that favor indexing.
- b. Recommend whether GAC should invest in Otunia actively or by indexing, and justify your recommendation based on the factors identified in Part a.

MSCI Emerging Frontier Markets Africa ex South Africa Index (USD)

The MSCI Emerging Frontier Markets Africa ex South Africa Index captures large and mid cap representation across 1 Emerging Market (EM) country and 12 Frontier Markets (FM) countries*. With 45 constituents, the index covers approximately 85% of the free float-adjusted market capitalization in each country.

For a complete description of the index methodology, please see [Index methodology - MSCI](#).

CUMULATIVE INDEX PERFORMANCE – GROSS RETURNS (USD) (AUG 2009 – AUG 2024)



ANNUAL PERFORMANCE (%)

Year	MSCI EFM Africa ex ZA	MSCI EFM	MSCI Frontier Markets
2023	7.53	10.27	12.17
2022	-21.71	-19.81	-26.05
2021	11.30	-2.01	20.09
2020	-4.30	18.39	1.65
2019	17.25	18.91	18.34
2018	-12.58	-14.28	-16.20
2017	22.81	37.62	32.32
2016	-7.43	11.40	3.16
2015	-19.20	-14.58	-14.07
2014	-5.43	-1.41	7.21
2013	18.67	-1.50	26.32
2012	41.49	18.35	9.25
2011	-30.36	-18.19	-18.38
2010	17.92	19.38	24.24

INDEX PERFORMANCE – GROSS RETURNS (%) (AUG 30, 2024)

	1 Mo	3 Mo	1 Yr	YTD	ANNUALIZED				Since May 31, 2002	FUNDAMENTALS (AUG 30, 2024)			
					3 Yr	5 Yr	10 Yr			Div Yld (%)	P/E	P/E Fwd	P/BV
MSCI EFM Africa ex ZA	3.17	7.62	3.02	-2.68	-6.10	-1.04	-3.38	7.94		3.83	10.25	na	1.90
MSCI EFM	1.65	6.08	15.44	9.86	-2.66	5.12	2.90	8.10		2.69	15.62	na	1.76
MSCI Frontier Markets	2.04	4.04	10.60	10.48	-2.24	3.19	1.38	6.87		4.18	10.59	na	1.52

INDEX RISK AND RETURN CHARACTERISTICS (AUG 30, 2024)

	Turnover (%) ¹	ANNUALIZED STD DEV (%) ²			SHARPE RATIO ^{2,3}			Since May 31, 2002	MAXIMUM DRAWDOWN	
		3 Yr	5 Yr	10 Yr	3 Yr	5 Yr	10 Yr		(%)	Period YYYY-MM-DD
MSCI EFM Africa ex ZA	21.24	14.97	17.61	16.34	-0.57	-0.09	-0.23	0.40	64.07	2008-04-21–2009-02-03
MSCI EFM	5.53	17.26	18.36	17.03	-0.27	0.24	0.15	0.40	63.90	2007-10-29–2008-10-27
MSCI Frontier Markets	33.31	13.47	16.52	14.23	-0.35	0.14	0.05	0.37	67.44	2008-01-15–2009-03-03

¹ Last 12 months

² Based on monthly gross returns data

³ Based on NY FED Overnight SOFR from Sep 1 2021 & on ICE LIBOR 1M prior that date

* EM countries include: Egypt. FM countries include: Burkina Faso, Benin, Guinea-Bissau, Ivory Coast, Kenya, Mauritius, Mali, Morocco, Niger, Senegal, Togo and Tunisia.

The MSCI Emerging Frontier Markets Africa ex South Africa Index was launched on Feb 07, 2008. Data prior to the launch date is back-tested test (i.e. calculations of how the index might have performed over that time period had the index existed). There are frequently material differences between back-tested performance and actual results. Past performance – whether actual or back-tested – is no indication or guarantee of future performance.

INDEX CHARACTERISTICS

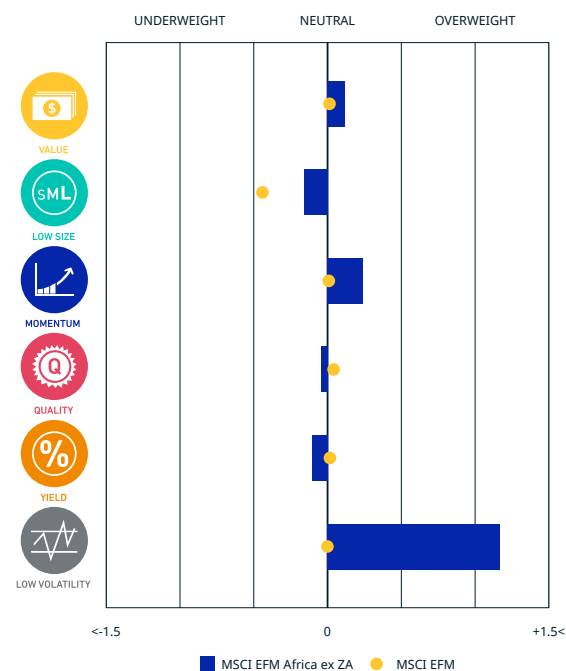
MSCI EFM Africa ex ZA	
Number of Constituents	45
Mkt Cap (USD Millions)	
Index	28,328.24
Largest	4,056.02
Smallest	75.09
Average	629.52
Median	334.05

TOP 10 CONSTITUENTS

	Country	Float Adj Mkt Cap (USD Billions)	Index Wt. (%)	Sector
COMMERCIAL INTL BANK	EG	4.06	14.32	Financials
ATTIJARIWABA BANK	MA	3.00	10.60	Financials
MCB GROUP	MU	2.06	7.29	Financials
MAROC TELECOM	MA	1.68	5.92	Comm Svcs
LAFARGEHOLCIM MAROC	MA	1.37	4.83	Materials
SAFARICOM	KE	1.35	4.78	Comm Svcs
TMG HOLDING	EG	1.13	3.98	Real Estate
SONATEL	SN	0.95	3.35	Comm Svcs
EQUITY GROUP HOLDINGS	KE	0.90	3.19	Financials
SOPEP MARSA MAROC	MA	0.86	3.05	Industrials
Total		17.37	61.31	

FACTORS - KEY EXPOSURES THAT DRIVE RISK AND RETURN

MSCI FACTOR BOX



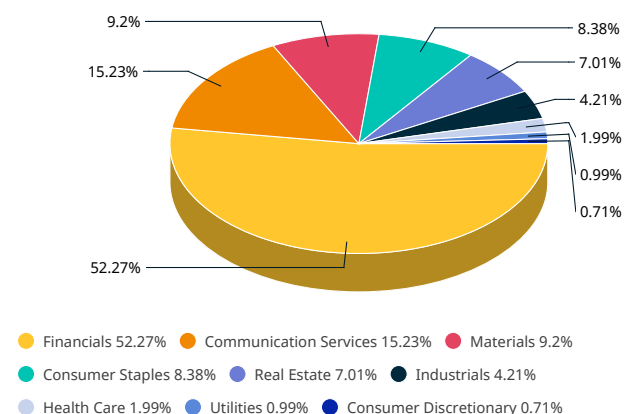
MSCI FaCS

- VALUE**
Relatively Inexpensive Stocks
- LOW SIZE**
Smaller Companies
- MOMENTUM**
Rising Stocks
- QUALITY**
Sound Balance Sheet Stocks
- YIELD**
Cash Flow Paid Out
- LOW VOLATILITY**
Lower Risk Stocks

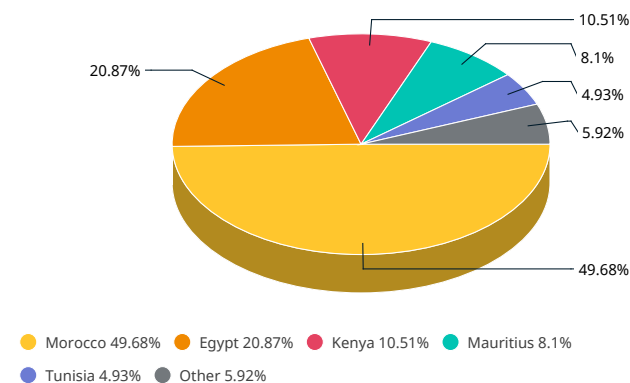
MSCI FaCS provides absolute factor exposures relative to a broad global index - MSCI ACWI IMI.

Neutral factor exposure (FaCS = 0) represents MSCI ACWI IMI.

SECTOR WEIGHTS



COUNTRY WEIGHTS



MSCI FACTOR BOX AND FaCS FRAMEWORK (Please refer to complete description of the MSCI FaCS methodology [here](#))

MSCI FaCS is a standard method for evaluating and reporting the Factor characteristics of equity portfolios. MSCI FaCS consists of Factor Groups (e.g. Value, Size, Momentum, Quality, Yield, and Volatility) that have been extensively documented in academic literature and validated by MSCI Research as key drivers of risk and return in equity portfolios. These Factor Groups are constructed by aggregating 16 factors (e.g. Book-to-Price, Earnings/Dividend Yields, LT Reversal, Leverage, Earnings Variability/Quality, Beta) from the latest Barra global equity factor risk model, GEMLT, designed to make fund comparisons transparent and intuitive for use. The MSCI Factor Box, which is powered by MSCI FaCS, provides a visualization designed to easily compare absolute exposures of funds/indexes and their benchmarks along 6 Factor Groups that have historically demonstrated excess market returns over the long run.

ABOUT MSCI

MSCI is a leading provider of critical decision support tools and services for the global investment community. With over 45 years of expertise in research, data and technology, we power better investment decisions by enabling clients to understand and analyze key drivers of risk and return and confidently build more effective portfolios. We create industry-leading research-enhanced solutions that clients use to gain insight into and improve transparency across the investment process. To learn more, please visit www.msci.com.

The information contained herein (the "Information") may not be reproduced or disseminated in whole or in part without prior written permission from MSCI. The Information may not be used to verify or correct other data, to create indexes, risk models, or analytics, or in connection with issuing, offering, sponsoring, managing or marketing any securities, portfolios, financial products or other investment vehicles. Historical data and analysis should not be taken as an indication or guarantee of any future performance, analysis, forecast or prediction. None of the Information or MSCI index or other product or service constitutes an offer to buy or sell, or a promotion or recommendation of, any security, financial instrument or product or trading strategy. Further, none of the Information or any MSCI index is intended to constitute investment advice or a recommendation to make (or refrain from making) any kind of investment decision and may not be relied on as such. The Information is provided "as is" and the user of the Information assumes the entire risk of any use it may make or permit to be made of the Information. NONE OF MSCI INC. OR ANY OF ITS SUBSIDIARIES OR ITS OR THEIR DIRECT OR INDIRECT SUPPLIERS OR ANY THIRD PARTY INVOLVED IN THE MAKING OR COMPILING OF THE INFORMATION (EACH, AN "MSCI PARTY") MAKES ANY WARRANTIES OR REPRESENTATIONS AND, TO THE MAXIMUM EXTENT PERMITTED BY LAW, EACH MSCI PARTY HEREBY EXPRESSLY DISCLAIMS ALL IMPLIED WARRANTIES, INCLUDING WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. WITHOUT LIMITING ANY OF THE FOREGOING AND TO THE MAXIMUM EXTENT PERMITTED BY LAW, IN NO EVENT SHALL ANY OF THE MSCI PARTIES HAVE ANY LIABILITY REGARDING ANY OF THE INFORMATION FOR ANY DIRECT, INDIRECT, SPECIAL, PUNITIVE, CONSEQUENTIAL (INCLUDING LOST PROFITS) OR ANY OTHER DAMAGES EVEN IF NOTIFIED OF THE POSSIBILITY OF SUCH DAMAGES. The foregoing shall not exclude or limit any liability that may not by applicable law be excluded or limited.

© 2024 MSCI Inc. All rights reserved.